

Human Label Distributions and Ambiguity-Aware Benchmarking of Bangla Handwritten Digits

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ABSTRACT

Bangla handwritten digit recognizers are evaluated against a single stored label per image and routinely exceed 99% accuracy, yet, to our knowledge, no Bangla benchmark records how people actually read its images. We collect human label distributions for the numeral subset of the Ekush corpus and use them for ambiguity-aware benchmarking. Ten blinded readers labelled 1,002 test images drawn by a pre-specified, partitioned stratified design that covers the whole test split, so corpus-level estimates are unbiased; nine passed pre-specified quality checks (Krippendorff's $\alpha = 0.947$). Genuine ambiguity is rare (0.66–3.38% of the test split, depending on the rule) and concentrated in the letterform pairs 5/6 and 1/9. Human consensus supports the stored label on 99.72% [99.54, 99.87] of the legible test images. Seed-ensembles of five architectures (three seeds each) reach 99.04–99.64%, and the 95% intervals of the four best overlap this human-support level. Only one of ten architecture pairs differs after Holm correction, and disagreement between seeds (1.02%) is close to disagreement between architectures (1.26%). Scored against the human distributions, models are over-confident exactly where readers disagree, and distributional metrics separate them more sharply than accuracy. Fine-tuning on human soft labels lowers negative log-likelihood against human labels by 0.030 nats without changing accuracy. The panel also confirms both confident-learning label-error candidates but finds two errors that confident learning missed (estimated recall 0.50). We release the soft labels, annotator-level responses and annotation tool.

1. Introduction

Bangla (Bengali) handwritten numeral recognition has a mature set of benchmarks—ISI [1], CMATERdb [2], BanglaLekha-Isolated [3], NumtaDB [4] and Ekush [5]—and deep networks report accuracies up to about 99.5% on them [6]. All of these benchmarks score a model against one stored label per image. Two questions are rarely asked. What is the correct answer for the images that remain, and do people agree on it? And are the reported differences between models real, or do they sit inside the run-to-run noise of training [7]? Some Bangla letterforms (for example 5 and 6, or 1 and 9) are close even in careful handwriting. A stored label records the digit a writer was *asked* to write, so on such images it can reward a model for matching that intent rather than for reading the image as people do. Label errors in test sets are known to distort model rankings [8].

Human label distributions make it possible to evaluate models against disagreement rather than one gold label [9–11]. Confident learning (CL) flags likely label errors [12], and deep ensembles estimate model uncertainty [13]. To our knowledge, no Bangla or other Indic handwriting benchmark has human label distributions, and none has human judgements collected so that they support *corpus-level* claims.

We build such distributions for the Ekush numerals and use them for ambiguity-aware benchmarking, in two stages (Fig. 1). Stage 1 trains and audits the models; its outputs—item selection, predictions, checkpoint hashes and every analysis threshold—are frozen and hashed (19 September

2026) before any person sees an image. Stage 2 collects blinded human label distributions on a partitioned stratified sample and re-evaluates the benchmark against them. Our contributions are: (i) a released human label-distribution dataset for 1,002 Ekush test digits, at annotator level, with the pre-specified analysis plan; (ii) a measurement of how much genuine ambiguity the benchmark contains, with annotator noise separated from image ambiguity; (iii) ambiguity-aware benchmarking: distributional metrics, a learned digit-confusion cost, agreement-conditioned accuracy and soft-label fine-tuning; (iv) a corpus-level estimate of human label support, directly comparable with test accuracy on the same split, showing that the benchmark is close to saturated; and (v) a human-validated audit of CL on a handwriting benchmark.

2. Related work

Bangla numeral recognition moved from hand-crafted features and multistage or feature-selection classifiers on the ISI and CMATERdb databases [1, 2] to deep convolutional networks [6], supported by larger collections such as BanglaLekha-Isolated [3], NumtaDB [4] and Ekush [5]. Results are often reported as single runs on one split, without seed variance or tests of label quality. In general benchmarks, CL [12] found test-set label errors that can reorder model rankings [8], and seed variance can rival published gaps [7], which calls for paired tests [14] with multiplicity correction [15]. CIFAR-10H showed that training on human label distributions improves robustness [9]; in NLP, annotator disagreement is increasingly treated as signal rather than

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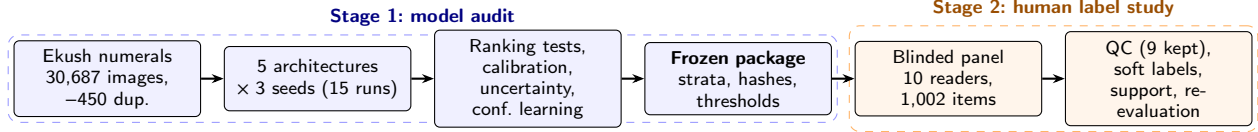


Figure 1: Two-stage study design. Everything that determines Stage 2 is frozen and hashed at the end of Stage 1, before annotation begins.

noise [10, 11], and releasing annotator-level labels is recommended [16]. Single-truth aggregation such as Dawid–Skene [17] instead attributes disagreement to annotator error. On the model side, temperature scaling [18] and deep ensembles [13] with an aleatoric/epistemic split [19] are standard tools for calibration and uncertainty.

3. Stage 1: model audit

Data. We use the ten-class numeral subset of Ekush [5] from a public redistribution of 28×28 grayscale images ([[mirror URL](#)]). After removing 450 exact duplicates, the 30,237 images are split once, stratified by class, into 24,189/3,024/3,024 train/validation/test images (seed 42). The split is fingerprinted with SHA-256. A difference-hash search confirmed by pixel correlation finds one train–test near-duplicate (same label) among 9,066 candidate pairs, so leakage is negligible (Supplementary Table S1).

Models. Xception [20], ResNet50V2 [21], MobileNetV2 [22], EfficientNetB0 [23] and a 0.29 M-parameter CNN are each trained with seeds 13, 42 and 2024. Every network takes the raw 28×28 image and resizes, replicates channels and scales it inside the model graph. The four backbones start from ImageNet weights; the custom CNN uses default Keras initialisation. We use Adam (learning rate 3×10^{-4} for the backbones, 10^{-3} for the custom CNN), batch size 128, random translation ($\pm 8\%$), rotation ($\pm 3\%$) and zoom ($\pm 8\%$), and at most 15 epochs with early stopping on the validation split (TensorFlow 2.20, mixed precision, one GPU). A *seed-ensemble* averages an architecture’s three seeds; the *deep ensemble* averages all 15 runs.

Analyses. We report Wilson intervals [24], prediction disagreement between runs, and paired comparisons of seed-ensembles (bootstrap CIs [25], exact McNemar tests [14] with Holm correction [15], bootstrap probability of ranking first). We also report calibration before and after temperature scaling [18], error detection with ensemble uncertainty [19], and CL on 5-fold out-of-fold predictions (train+validation) or ensemble predictions (test).

Findings. All seed-ensembles lie between 99.04% and 99.64% (Table 1). ResNet50V2 ranks first in 84% of bootstrap resamples, but its rank interval (1–2) overlaps those of MobileNetV2 (1–3) and Xception (1–4). Xception’s three seeds span 0.53 percentage points (pp), more than its gap to ResNet50V2. Two runs of the same architecture disagree on 1.02% of test images, against 1.26% for different architectures (Fig. 2). Only ResNet50V2 versus EfficientNetB0 survives Holm correction (+0.60 pp [0.23, 0.96], $p_{\text{Holm}} = 0.014$; Supplementary Table S3), and the 0.29 M-parameter

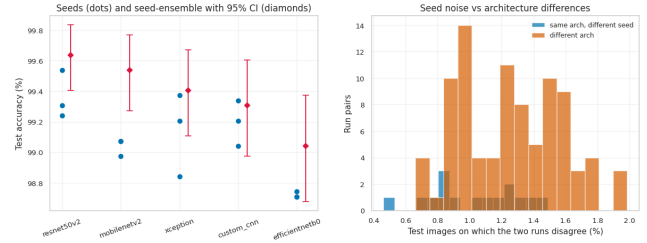


Figure 2: Left: individual seeds (dots) and seed-ensembles with 95% CIs (diamonds). Right: prediction disagreement between pairs of runs; same-architecture pairs (seed noise) overlap with different-architecture pairs.

CNN is not significantly different from the 20.9 M-parameter Xception ($p_{\text{Holm}} = 1.0$). All models are well calibrated: fitted temperatures are 0.90–1.13, and scaling changes negative log-likelihood (NLL) by about 0.001 or less (Supplementary Table S4). The 15-run ensemble makes 10 errors, and uncertainty scores detect them with AUROC 0.88–0.93 (Supplementary Table S5). Errors are mostly idiosyncratic: only 11 images are misclassified by at least half of the runs. CL flags 40 of 27,213 train+validation images (0.15%) and 2 of 3,024 test images (0.07%) (Supplementary Table S6).

4. Stage 2: human label study

Sampling. A sample drawn only from model-ambiguous images would bias every corpus-level estimate. We therefore partition the test split into four disjoint strata that cover it completely, ordered by an ensemble ambiguity score (rank-average of predictive entropy, aleatoric entropy, top-2 margin and variation ratio): a census of CL candidates (c1), a census of the next 600 most ambiguous images (high), 200 sampled from the following 600 (mid), and 200 from the remaining 1,822 (low) (Table 2). Corpus quantities use the stratified estimator $\hat{\theta} = \sum_s (N_s/N) \hat{\theta}_s$ with a within-stratum bootstrap ($B = 5000$).

Instrument and panel. Annotators used an offline bilingual (Bangla/English) HTML tool. It was blinded (opaque item IDs, no stored label, no model output) and used a per-annotator random order. For each image they chose one of the ten digits, *illegible* or *not a digit*, and a confidence level. Fifty easy items were repeated as attention checks. Ten Bangla-speaking volunteers gave informed consent and were identified only by study IDs. All thresholds were pre-specified and hash-frozen on 19 September 2026, before annotation began (SHA-256 e2b407b9...; Supplementary Table S24). One annotator exceeded the pre-specified limit

Table 1

Models under both stages. Left: full test split ($n = 3024$); seed mean $\pm$ sd over three seeds, seed-ensemble accuracy with Wilson 95% CI, bootstrap P (rank 1), and NLL (mean over seeds). Right: annotated legible items, scored against the human panel (unweighted means over the ambiguity-enriched sample, not corpus rates); accuracy against the human mode on all items ($n = 999$) and on ambiguous items ($n = 20$; 95% CIs span about ± 20 pp, Supplementary Table S18), Jensen–Shannon divergence (JSD) and confusion-weighted cost against the human distribution (lower is better). Best value per column in bold.

Model	Params (M)	Full test split vs. Ekush labels					Annotated items vs. human labels			
		Seed mean $\pm$ sd (%)	Seed-ensemble (%) [95% CI]	P (rank 1)	NLL		Mode acc. (%)	Ambig. acc. (%)	JSD	Conf. cost
ResNet50V2	23.59	99.36 $\pm$ 0.16	99.64 [99.35, 99.80]	0.84	0.025		98.7	80	0.012	0.020
MobileNetV2	2.27	99.04 $\pm$ 0.06	99.54 [99.22, 99.72]	0.22	0.034		98.4	70	0.019	0.031
Xception	20.88	99.14 $\pm$ 0.27	99.40 [99.06, 99.62]	0.04	0.033		98.2	70	0.016	0.025
Custom CNN	0.29	99.20 $\pm$ 0.15	99.31 [98.94, 99.55]	0.00	0.037		97.8	65	0.023	0.039
EfficientNetB0	4.06	98.72 $\pm$ 0.02	99.04 [98.63, 99.33]	0.00	0.047		96.9	65	0.025	0.036
Deep ensemble (15 runs)	–	–	99.67 [99.39, 99.82]	–	–		98.8	75	0.017	0.030

on responses faster than 700 ms (21% of answers) and was excluded, leaving nine annotators and 9,018 judgements. The retained annotators had repeat consistency ≥ 0.98 and control accuracy ≥ 0.98 (Supplementary Table S8).

Soft labels and taxonomy. For item i , $q_i(c)$ is the share of digit votes for class c . Pre-specified rules, applied in order, label an item *insufficient* (< 5 digit votes), *illegible* (illegible share ≥ 0.5), a *suspected mislabel* (mode $\neq$ Ekush label with share ≥ 0.8), *unambiguous* (at most one dissenting vote) or *ambiguous*. To separate annotator noise from ambiguity, each annotator’s error rate e_a is measured on clear control items (Jeffreys prior). An item counts as noise-consistent when its dissent is plausible under a Poisson-binomial error count. A one-coin posterior $\pi_i(c) \propto \prod_a (1 - e_a)^{[r_{ai}=c]} (e_a/9)^{[r_{ai} \neq c]}$ weights reliable annotators more. We do not use Dawid–Skene [17], which on ambiguous items attributes genuine disagreement to annotator error.

Estimands and evaluation. The primary estimand A_{cons} is the corpus share of items whose panel majority reads the Ekush label (ties count fractionally). A_{wmv} is the mean posterior $\pi_i(y_i)$, and $C_{\text{annot}} = \mathbb{E}[\max_c q_i(c)]$ is agreement with a single random reader. A model significantly *above* A_{cons} would be matching labels that people do not read. Models are also scored against the human mode, and against q_i with Jensen–Shannon divergence (JSD) [26], total variation, soft Brier, cross-entropy and a confusion-weighted cost $\sum_{c,k} q_i(c) D_{ck} p_i(k)$. The ground cost D is learned from human co-voting and cross-fitted over two halves of the items. To test for annotator sub-groups, the residual of a rank-1 fit to the agreement matrix is compared with a parametric null in which annotators differ only in skill; we check this test’s size and power on matched synthetic panels.

5. Results

Agreement and ambiguity. Agreement is high: Krippendorff’s α [27] is 0.947 [0.938, 0.956], and lowest on high (0.920), which is the stratum the study targets (Table 2). Of the 1,002 items, 975 are unambiguous, 20 ambiguous, 4 suspected mislabels and 3 insufficient. No item meets the illegible rule, but the three insufficient items were mostly marked illegible or not a digit (0–4 digit votes out of 9; about 0.10% of the corpus). Weighted to the corpus, 0.66%

Table 2

Partitioned design, agreement (Krippendorff’s α) and item taxonomy by stratum. Pop. = stratum size in the test split; Ann. = annotated items.

Stratum	Pop.	Ann.	α	Unamb.	Amb.	Mislab.	Insuff.
c1	2	2	1.000	0	0	2	0
high	600	600	0.920	575	20	2	3
mid	600	200	0.981	200	0	0	0
low	1,822	200	0.988	200	0	0	0
All	3,024	1,002	0.947	975	20	4	3

of the test split is ambiguous under the primary rule and 3.38% if a single dissenting vote suffices. Annotator error rates on clear items are so low ($e_a = 0.25$ – 0.75%) that the noise-consistency rule coincides with unanimity, so no disagreement is explained away as noise. Model and human uncertainty are only weakly related: 575 of the 600 images the ensemble found most ambiguous are unambiguous to people (Spearman $\rho = 0.20$ between ensemble aleatoric entropy and human entropy). Human disagreement concentrates in a few letterform pairs (Fig. 3a): among ambiguous items, the two most-voted digits are 5/6 in 8 cases and 1/9 in 3. The cross-fitted confusion cost recovers the same structure, with cheapest pairs 5 $\leftrightarrow$ 6 (0.02), 1 $\leftrightarrow$ 9 (0.35), 3 $\leftrightarrow$ 6 (0.51) and 7 $\leftrightarrow$ 9 (0.57) (Supplementary Fig. S9).

Label support. The panel majority reads the Ekush label on an estimated $A_{\text{cons}} = 99.72\%$ [99.54, 99.87] of the legible test images (Table 3). The reliability-weighted estimate is almost identical, and every mid and low item supports its label. The 95% intervals for ResNet50V2, MobileNetV2, Xception, the custom CNN and the deep ensemble all overlap the A_{cons} interval (for the custom CNN only marginally); only EfficientNetB0 lies below it (Fig. 4). No model is significantly above human support, so we find no evidence that the top models fit label idiosyncrasies. The best models already agree with the stored labels about as often as a consensus of careful readers does.

Label errors and CL. Both test-set CL candidates are confirmed: all nine readers agree on a different digit, which matches CL’s suggested label. The panel finds two more unanimous errors in high, both Ekush “1” read as “9” (Fig. 3b). No errors appear in the sampled mid and low strata,

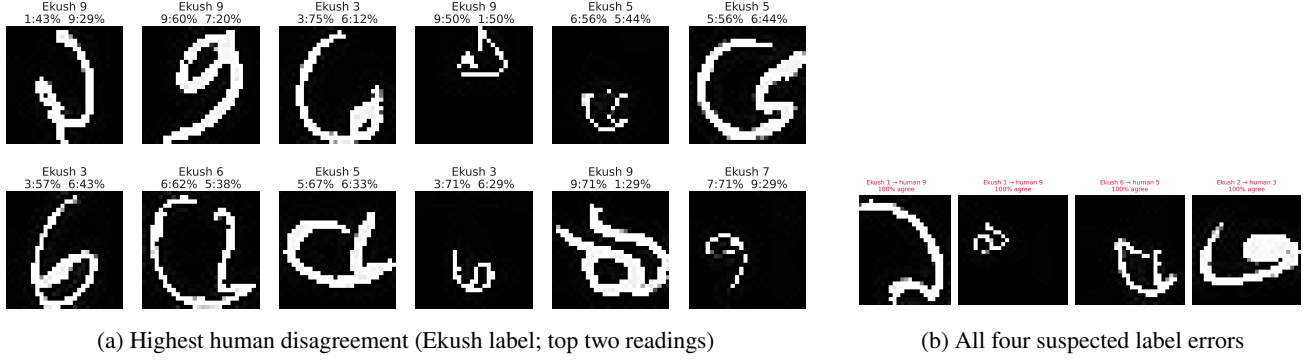


Figure 3: (a) The twelve items with the highest human label entropy. (b) The four suspected mislabels: stored label, human reading and panel agreement. The two right-hand items in (b) are the confident-learning candidates; the two 1→9 errors were missed by confident learning.

Table 3

Corpus-level estimates (stratified, within-stratum bootstrap 95% CI; 999 legible items) and the human validation of confident learning (CL).

Quantity	Estimate [95% CI]
A_{cons} : majority reads Ekush label	99.72% [99.54, 99.87]
A_{wmv} : reliability-weighted support	99.73% [99.56, 99.87]
C_{annot} : agreement with one reader	99.47% [99.30, 99.63]
Ambiguous share (primary / any dissent)	0.66% / 3.38%
Unreadable (insufficient digit votes)	0.10%
Estimated label errors in test split	≈4 (0.13%)
CL precision (confirmed / candidates)	2/2 [0.34, 1.00]
CL recall (estimated)	0.50 [0.29, 1.00]

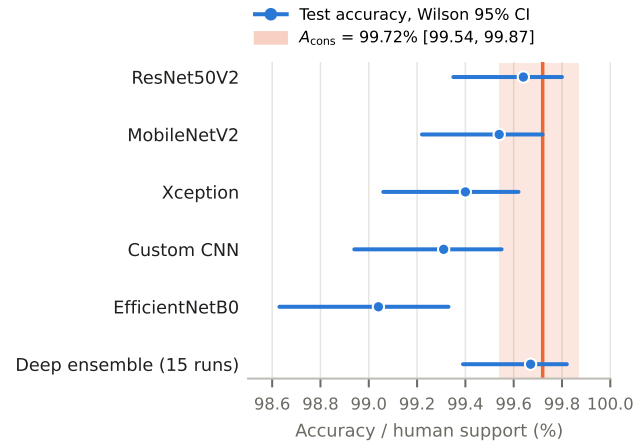


Figure 4: Full test-split accuracy of each seed-ensemble and of the deep ensemble (Wilson 95% CI) against human label support A_{cons} (line: estimate; band: stratified bootstrap 95% CI).

so CL's estimated recall is 0.50 and the test label-error rate is about 0.13%. The missed errors suggest a blind spot of CL: where a letterform pair is systematically confusable, the models may not be confidently wrong about it.

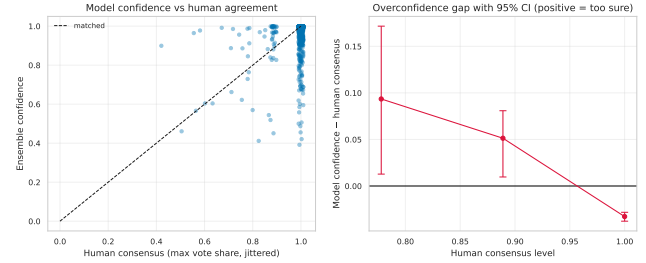


Figure 5: Left: ensemble confidence against human consensus (jittered). Right: over-confidence gap (model confidence – human consensus) with 95% CIs; models are over-confident exactly where readers disagree.

Rankings and over-confidence. On the annotated items the point ranking is unchanged in every scoring regime apart from ties on ambiguous items (Supplementary Table S15), but only 1–2 of 10 pairs are significant after Holm correction, and none on the 20 ambiguous items, where ResNet50V2 matches the human mode on 16. Ambiguous items are much harder for every model: 65–80% accuracy against the human mode, against 97.6–99.5% on unambiguous items. The ranking is nearly the same on both subsets (Spearman $\rho = 0.95$), and parameter count does not predict accuracy on ambiguous items. Hard-label calibration hides a mismatch with people. Where all readers agree, the ensemble is slightly under-confident (gap -0.033). Where one reader dissents it is over-confident by $+0.051$ [0.010, 0.081], and where two dissent by $+0.094$ [0.013, 0.172] (Fig. 5; Supplementary Table S16).

Distributional metrics. Against q_i , ResNet50V2 is best on JSD, total variation and confusion cost, with bootstrap $P(\text{best}) \geq 0.98$ on each (Table 1; Supplementary Table S17). The deep ensemble ties with it on KL divergence and cross-entropy. The metrics mostly agree (Kendall's $W = 0.89$ [28]), but every model's rank depends on the metric chosen.

Annotator structure and panel size. Annotators are not exchangeable (within-item permutation test, $p = 0.001$), but the skill-aware test finds no structure beyond skill differences

Table 4

Fine-tuning target (Xception; out-of-fold, $n = 999$, mean over 3 seeds). Bottom: paired bootstrap difference, soft – hard (Ekush), with 95% CI.

Target	Acc.	Soft Brier	NLL (q)	ECE
Hard (Ekush label)	0.981	0.0278	0.1448	0.0144
Hard (human mode)	0.980	0.0264	0.1317	0.0127
Soft (human q)	0.980	0.0251	0.1105	0.0091
Δ soft – Ekush	+0.001 [−.002, .004]	−0.0030 [−.0064, −.0001]	−0.0295 [−.0434, −.0177]	−0.0041 [−.0108, .0030]

($p = 0.325$). On matched synthetic panels this test holds its size (rejection rate 0.02 at $\alpha = 0.05$) and detects moderate shared reading conventions with power 0.93 (Supplementary Table S20). In random sub-panels, label support and agreement reach their full-panel values (within 0.5 and 5 points) with two annotators, but the estimated ambiguous share keeps rising with panel size, from 0.00% with two annotators to 0.66% with nine (Supplementary Table S21). Estimating label support needs few readers; characterising ambiguity needs larger panels.

Soft-label fine-tuning. The seed-13 Xception checkpoint is fine-tuned on the legible annotated items with three targets, using 5-fold cross-validation stratified by human mode and repeated over three seeds (Adam, learning rate 10^{-4} , batch 32). Because these test-split images are used for fine-tuning (always out-of-fold), the resulting numbers are not benchmark accuracies. Accuracy is unchanged (Table 4). Soft targets lower NLL against q_i by 0.030 [0.018, 0.043] relative to hard Ekush labels and by 0.018 [0.010, 0.028] relative to the hard human mode, and they also lower the soft Brier score. The gain therefore comes from the distribution itself, not only from correcting four labels. Expected calibration error (ECE) [29] falls on average, but its paired CI includes zero.

6. Discussion and limitations

The Ekush numeral labels are sound. About 99.7% are supported by human consensus, only about 0.13% appear wrong, and the top models already sit at the human-support level. On this split, accuracy differences of a few tenths of a point are the same size as seed noise, binomial uncertainty and the uncertainty of the human reference. Future Ekush results should therefore report several seeds, paired tests with multiplicity correction, and accuracy on unambiguous items together with distributional metrics against the released soft labels. The remaining difficulty is a small set of letterforms that sit between two digits. On these images no single answer is correct, and models are over-confident. Model uncertainty is a poor stand-in for human ambiguity: 96% of model-ambiguous images were clear to people. CL is precise but misses errors in exactly the confusable classes.

Several limitations apply. We study a single benchmark, Ekush; other Bangla digit sets were collected differently and may contain more or less ambiguity. The comparison with A_{cons} rests on overlapping intervals rather than a paired test, because A_{cons} is estimated on the annotated sample while

accuracy uses the full split. Images are 28×28 , so some ambiguity may be a resolution artefact. We use a single split. Item selection depends on the models, although the partitioned design keeps corpus estimates unbiased. The analysis plan was hash-frozen before annotation but not lodged in a public registry. The ambiguous subset (20 items) and the CL stratum (2 items) are small. Each item has only nine votes. The one-coin noise model assumes an annotator's error rate is the same on hard and clear images. Finally, the readers are not the writers and may not share every regional letterform convention.

7. Conclusion

Human label distributions change how the Ekush Bangla numeral benchmark should be read. A blinded, corpus-representative panel with a pre-specified analysis plan shows that the stored labels are sound and that genuine ambiguity is rare but structured, concentrated in a few letterform pairs. On those images, models are over-confident, CL is incomplete, and accuracy against one label says little. Ambiguity-aware evaluation separates the models more clearly than accuracy, while most accuracy-based rankings are not resolved once seed noise and multiplicity are taken into account. The released distributions let future work on Bangla handwriting be evaluated against how people actually read it.

CRedit authorship contribution statement

Jakaria: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **[Co-author name]:** [CRedit roles].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

Annotators were Bangla-speaking volunteers who were told the purpose and procedure of the study and gave informed consent before annotating. No personally identifying information was collected; annotators are identified only by study IDs.

Data availability

Human soft labels, annotator-level responses (10,520 rows), annotator noise rates, the learned 10×10 confusion

cost, the pre-specified analysis plan (preregistration.json), the annotation tool and a datasheet are available at <https://github.com/Yeakob/Bangla-Handwritten-digit> (labels under CC BY 4.0; the annotation tool embeds Ekush images, which are redistributed for research use only, as permitted by Ekush [5]). Supplementary material gives full methods, all tables and all figures.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to draft and format the manuscript text and tables from the authors' analysis outputs. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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